A3

1.1. Mixture of Bernoulli

1. Mixture of K Multivariate Bernoulli Distributions $p(\mathbf{x} \mid \Theta)$

The parameter set is $\Theta = \{\boldsymbol{\mu}_k, \pi_k \mid k = 1, \ldots, K\}$, where $\boldsymbol{\mu}_k$ is the mean vector for the k-th component, and π_k is the mixing coefficient satisfying $\sum_{k=1}^{K} \pi_k = 1$. The single component distribution is given as:

$$p(\mathbf{x} \mid \boldsymbol{\mu}_k) = \prod_{j=1}^{m} \mu_k[j]^{x[j]} (1 - \mu_k[j])^{1-x[j]}$$

Since this is a mixture model, the total probability is the weighted sum of the component distributions, weighted by the mixing coefficients π_k. Thus, the mixture distribution is:

$$p(\mathbf{x} \mid \Theta) = \sum_{k=1}^{K} \pi_k p(\mathbf{x} \mid \boldsymbol{\mu}_k) = \sum_{k=1}^{K} \pi_k \prod_{j=1}^{m} \mu_k[j]^{x[j]} (1 - \mu_k[j])^{1-x[j]}$$

2. Log-Likelihood $L(\Theta)$ for Dataset $X = \{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\}$

Assume the data points are independent. The likelihood is the product of the individual probabilities:

$$p(X \mid \Theta) = \prod_{i=1}^{n} p(\mathbf{x}^{(i)} \mid \Theta)$$

Substituting the mixture distribution from above:

$$p(X \mid \Theta) = \prod_{i=1}^{n} \sum_{k=1}^{K} \pi_k \prod_{j=1}^{m} \mu_k[j]^{x^{(i)}[j]} (1 - \mu_k[j])^{1-x^{(i)}[j]}$$

The log-likelihood $L(\Theta)$ is the logarithm of this likelihood:

$$L(\Theta) = \log p(X \mid \Theta) = \log \left(\prod_{i=1}^{n} \sum_{k=1}^{K} \pi_k \prod_{j=1}^{m} \mu_k[j]^{x^{(i)}[j]} (1 - \mu_k[j])^{1-x^{(i)}[j]} \right) = \sum_{i=1}^{n} \log \left(\sum_{k=1}^{K} \pi_k \prod_{j=1}^{m} \mu_k[j]^{x^{(i)}[j]} (1 - \mu_k[j])^{1-x^{(i)}[j]} \right)$$

3. Conditional Distribution $p(\mathbf{x}_1 \mid \mathbf{x}_2; \Theta)$

The conditional probability is:

$$p(\mathbf{x}_1 \mid \mathbf{x}_2; \Theta) = \frac{p(\mathbf{x}_1, \mathbf{x}_2 \mid \Theta)}{p(\mathbf{x}_2 \mid \Theta)}$$

Since $\mathbf{x} = [\mathbf{x}_1, \mathbf{x}_2]$, the joint distribution is:

$$p(\mathbf{x}_1, \mathbf{x}_2 \mid \Theta) = p(\mathbf{x} \mid \Theta) = \sum_{k=1}^{K} \pi_k \prod_{j=1}^{m} \mu_k[j]^{x[j]} (1 - \mu_k[j])^{1-x[j]}$$

Because the components of $\mathbf{x}$ are independent given k (as the product is over j), split the product over the dimensions of $\mathbf{x}_1$ (first $m - d$ components) and $\mathbf{x}_2$ (last d components). Assume indices 1 to $m - d$ correspond to $\mathbf{x}_1$ and $m - d + 1$ to m correspond to $\mathbf{x}_2$:

$$p(\mathbf{x} \mid \boldsymbol{\mu}_k) = \prod_{j=1}^{m-d} \mu_k[j]^{x_1[j]} (1 - \mu_k[j])^{1-x_1[j]} \cdot \prod_{j=m-d+1}^{m} \mu_k[j]^{x_2[j]} (1 - \mu_k[j])^{1-x_2[j]}$$

Define:

- $p(\mathbf{x}_1 \mid \boldsymbol{\mu}_k) = \prod_{j=1}^{m-d} \mu_k[j]^{x_1[j]} (1 - \mu_k[j])^{1-x_1[j]}$
- $p(\mathbf{x}_2 \mid \boldsymbol{\mu}_k) = \prod_{j=m-d+1}^{m} \mu_k[j]^{x_2[j]} (1 - \mu_k[j])^{1-x_2[j]}$

So:

$$p(\mathbf{x} \mid \boldsymbol{\mu}_k) = p(\mathbf{x}_1 \mid \boldsymbol{\mu}_k) p(\mathbf{x}_2 \mid \boldsymbol{\mu}_k)$$

Thus:

$$p(\mathbf{x}_1, \mathbf{x}_2 \mid \Theta) = \sum_{k=1}^{K} \pi_k p(\mathbf{x}_1 \mid \boldsymbol{\mu}_k) p(\mathbf{x}_2 \mid \boldsymbol{\mu}_k)$$

The marginal probability is:

$$p(\mathbf{x}_2 \mid \Theta) = \sum_{\mathbf{x}_1} p(\mathbf{x}_1, \mathbf{x}_2 \mid \Theta) = \sum_{\mathbf{x}_1} \sum_{k=1}^{K} \pi_k p(\mathbf{x}_1 \mid \boldsymbol{\mu}_k) p(\mathbf{x}_2 \mid \boldsymbol{\mu}_k)$$

Since $\sum_{\mathbf{x}_1} p(\mathbf{x}_1 \mid \boldsymbol{\mu}_k) = 1$ (it's a probability distribution over $\mathbf{x}_1$):

$$p(\mathbf{x}_2 \mid \Theta) = \sum_{k=1}^{K} \pi_k p(\mathbf{x}_2 \mid \boldsymbol{\mu}_k)$$

Now, the conditional distribution becomes:

$$p(\mathbf{x}_1 \mid \mathbf{x}_2; \Theta) = \frac{\sum_{k=1}^{K} \pi_k p(\mathbf{x}_1 \mid \boldsymbol{\mu}_k) p(\mathbf{x}_2 \mid \boldsymbol{\mu}_k)}{\sum_{k'=1}^{K} \pi_{k'} p(\mathbf{x}_2 \mid \boldsymbol{\mu}_{k'})} = \sum_{k=1}^{K} \left(\frac{\pi_k p(\mathbf{x}_2 \mid \boldsymbol{\mu}_k)}{\sum_{k'} \pi_{k'} p(\mathbf{x}_2 \mid \boldsymbol{\mu}_{k'})} \right) p(\mathbf{x}_1 \mid \boldsymbol{\mu}_k)$$

Define the mixing coefficients:

$$\gamma_k(\mathbf{x}_2) = \frac{\pi_k p(\mathbf{x}_2 \mid \boldsymbol{\mu}_k)}{\sum_{k'=1}^{K} \pi_{k'} p(\mathbf{x}_2 \mid \boldsymbol{\mu}_{k'})} = \frac{\pi_k \prod_{j=m-d+1}^{m} \mu_k[j]^{x_2[j]} (1 - \mu_k[j])^{1 - x_2[j]}}{\sum_{k'} \pi_{k'} \prod_{j=m-d+1}^{m} \mu_{k'}[j]^{x_2[j]} (1 - \mu_{k'}[j])^{1 - x_2[j]}}$$

The component densities are:

$$p(\mathbf{x}_1 \mid \boldsymbol{\mu}_k) = \prod_{j=1}^{m-d} \mu_k[j]^{x_1[j]} (1 - \mu_k[j])^{1 - x_1[j]}$$

Thus:

$$p(\mathbf{x}_1 \mid \mathbf{x}_2; \Theta) = \sum_{k=1}^{K} \gamma_k(\mathbf{x}_2) \prod_{j=1}^{m-d} \mu_k[j]^{x_1[j]} (1 - \mu_k[j])^{1 - x_1[j]}$$

This is a mixture of K multivariate Bernoulli distributions over $\mathbf{x}_1$, with mixing coefficients $\gamma_k(\mathbf{x}_2)$ that depend on $\mathbf{x}_2$ and component densities parameterized by the first $m - d$ elements of $\boldsymbol{\mu}_k$.

4. Latent Variable Distributions

$p(\mathbf{z}^{(i)} \mid \pi)$:

Since $\mathbf{z}^{(i)}$ indicates the component assignment, if $z_k^{(i)} = 1$, the probability is π_k. Formally, as a categorical distribution:

$$p(\mathbf{z}^{(i)} \mid \pi) = \prod_{k=1}^{K} \pi_k^{z_k^{(i)}}$$

This evaluates to π_k when $z_k^{(i)} = 1$ and others are 0, since $\pi_k^1 \cdot \prod_{k' \neq k} \pi_{k'}^0 = \pi_k$.

$p(\mathbf{x}^{(i)} \mid \mathbf{z}^{(i)}, \Theta)$:

Given $\mathbf{z}^{(i)}$, $\mathbf{x}^{(i)}$ is generated by the component where $z_k^{(i)} = 1$:

$$p(\mathbf{x}^{(i)} \mid \mathbf{z}^{(i)}, \Theta) = \prod_{k=1}^{K} \left[p(\mathbf{x}^{(i)} \mid \boldsymbol{\mu}_k) \right]^{z_k^{(i)}} = \prod_{k=1}^{K} \left[\prod_{j=1}^{m} \mu_k[j]^{x^{(i)}[j]} (1 - \mu_k[j])^{1 - x^{(i)}[j]} \right]^{z_k^{(i)}}$$

If $z_k^{(i)} = 1$, this becomes $p(\mathbf{x}^{(i)} \mid \boldsymbol{\mu}_k)$.

$p(Z, X \mid \Theta)$:

Assuming independence across data points:

$$p(Z, X \mid \Theta) = \prod_{i=1}^{n} p(\mathbf{z}^{(i)}, \mathbf{x}^{(i)} \mid \Theta) = \prod_{i=1}^{n} p(\mathbf{z}^{(i)} \mid \pi) p(\mathbf{x}^{(i)} \mid \mathbf{z}^{(i)}, \Theta)$$

So:

$$p(Z, X \mid \Theta) = \prod_{i=1}^{n} \left[\prod_{k=1}^{K} \pi_k^{z_k^{(i)}} \right] \left[\prod_{k=1}^{K} \left(\prod_{j=1}^{m} \mu_k[j]^{x^{(i)}[j]} (1 - \mu_k[j])^{1 - x^{(i)}[j]} \right)^{z_k^{(i)}} \right]$$

5. Expression for $\phi(z_k^{(i)}) = \mathbb{E}[z_k^{(i)} \mid \mathbf{x}^{(i)}, \Theta]$

Since $z_k^{(i)}$ is binary, $\mathbb{E}[z_k^{(i)} \mid \mathbf{x}^{(i)}, \Theta] = p(z_k^{(i)} = 1 \mid \mathbf{x}^{(i)}, \Theta)$. Using Bayes' theorem:

$$p(z_k^{(i)} = 1 \mid \mathbf{x}^{(i)}, \Theta) = \frac{p(\mathbf{x}^{(i)} \mid z_k^{(i)} = 1, \Theta) p(z_k^{(i)} = 1 \mid \Theta)}{p(\mathbf{x}^{(i)} \mid \Theta)}$$

- $p(z_k^{(i)} = 1 \mid \Theta) = \pi_k$
- $p(\mathbf{x}^{(i)} \mid z_k^{(i)} = 1, \Theta) = p(\mathbf{x}^{(i)} \mid \boldsymbol{\mu}_k) = \prod_{j=1}^{m} \mu_k[j]^{x^{(i)}[j]} (1 - \mu_k[j])^{1 - x^{(i)}[j]}$
- $p(\mathbf{x}^{(i)} \mid \Theta) = \sum_{k'=1}^{K} \pi_{k'} p(\mathbf{x}^{(i)} \mid \boldsymbol{\mu}_{k'})$

Thus:

$$\phi(z_k^{(i)}) = \frac{\pi_k \prod_{j=1}^{m} \mu_k[j]^{x^{(i)}[j]} (1 - \mu_k[j])^{1 - x^{(i)}[j]}}{\sum_{k'=1}^{K} \pi_{k'} \prod_{j=1}^{m} \mu_{k'}[j]^{x^{(i)}[j]} (1 - \mu_{k'}[j])^{1 - x^{(i)}[j]}}$$

6. E-Step Expression $\mathbb{E}[\ln p(Z, X \mid \Theta) \mid X, \Theta]$

$$\ln p(Z, X \mid \Theta) = \sum_{i=1}^{n} \left[\ln p(\mathbf{z}^{(i)} \mid \pi) + \ln p(\mathbf{x}^{(i)} \mid \mathbf{z}^{(i)}, \Theta) \right]$$

- $\ln p(\mathbf{z}^{(i)} \mid \pi) = \sum_{k=1}^{K} z_k^{(i)} \ln \pi_k$
- $\ln p(\mathbf{x}^{(i)} \mid \mathbf{z}^{(i)}, \Theta) = \sum_{k=1}^{K} z_k^{(i)} \ln p(\mathbf{x}^{(i)} \mid \boldsymbol{\mu}_k)$

So:

$$\mathbb{E}[\ln p(Z, X \mid \Theta) \mid X, \Theta] = \sum_{i=1}^{n} \mathbb{E}_{\mathbf{z}^{(i)} \mid \mathbf{x}^{(i)}, \Theta} \left[\sum_{k=1}^{K} z_k^{(i)} \ln \pi_k + \sum_{k=1}^{K} z_k^{(i)} \ln p(\mathbf{x}^{(i)} \mid \boldsymbol{\mu}_k) \right]$$

Since $\mathbb{E}[z_k^{(i)} \mid \mathbf{x}^{(i)}, \Theta] = \phi(z_k^{(i)})$:

$$\mathbb{E}[\ln p(Z, X \mid \Theta) \mid X, \Theta] = \sum_{i=1}^{n} \sum_{k=1}^{K} \phi(z_k^{(i)}) \left[\ln \pi_k + \ln p(\mathbf{x}^{(i)} \mid \boldsymbol{\mu}_k) \right]$$

Where:

$$\ln p(\mathbf{x}^{(i)} \mid \boldsymbol{\mu}_k) = \sum_{j=1}^{m} \left[x^{(i)}[j] \ln \mu_k[j] + (1 - x^{(i)}[j]) \ln (1 - \mu_k[j]) \right]$$

7. Maximizing Parameters $\overline{\boldsymbol{\mu}}_k$ and $\bar{\pi}_k$

Maximize the E-step expression with respect to $\boldsymbol{\mu}_k$ and π_k, subject to $\sum_{k=1}^{K} \pi_k = 1$.

For π_k:

The term involving π_k is:

$$\sum_{i=1}^{n} \sum_{k=1}^{K} \phi(z_k^{(i)}) \ln \pi_k = \sum_{k=1}^{K} N_k \ln \pi_k, \quad N_k = \sum_{i=1}^{n} \phi(z_k^{(i)})$$

Maximize subject to $\sum_k \pi_k = 1$ using a Lagrange multiplier:

$$\mathcal{L} = \sum_{k=1}^{K} N_k \ln \pi_k + \lambda(1 - \sum_{k=1}^{K} \pi_k)$$

$$\frac{\partial \mathcal{L}}{\partial \pi_k} = \frac{N_k}{\pi_k} - \lambda = 0 \implies \pi_k = \frac{N_k}{\lambda}$$

$$\sum_k \pi_k = 1 \implies \sum_k \frac{N_k}{\lambda} = 1 \implies \lambda = \sum_k N_k = n$$

Thus:

$$\bar{\pi}_k = \frac{N_k}{\sum_{k=1}^{K} N_k}$$

Since $\sum_k N_k = \sum_i \sum_k \phi(z_k^{(i)}) = \sum_i 1 = n$, this is $\frac{N_k}{n}$, but the form matches the problem statement.

For $\boldsymbol{\mu}_k$:

Maximize:

$$\sum_{i=1}^{n} \sum_{k=1}^{K} \phi(z_k^{(i)}) \ln p(\mathbf{x}^{(i)} \mid \boldsymbol{\mu}_k)$$

For each k:

$$\sum_{i=1}^{n} \phi(z_k^{(i)}) \sum_{j=1}^{m} \left[x^{(i)}[j] \ln \mu_k[j] + (1 - x^{(i)}[j]) \ln(1 - \mu_k[j]) \right]$$

Per dimension j:

$$\sum_{i=1}^{n} \phi(z_k^{(i)}) \left[x^{(i)}[j] \ln \mu_k[j] + (1 - x^{(i)}[j]) \ln(1 - \mu_k[j]) \right]$$

Derivative with respect to $\mu_k[j]$:

$$\sum_{i=1}^{n} \phi(z_k^{(i)}) \left[\frac{x^{(i)}[j]}{\mu_k[j]} - \frac{1 - x^{(i)}[j]}{1 - \mu_k[j]} \right] = 0$$

$$\sum_{i=1}^{n} \phi(z_k^{(i)}) x^{(i)}[j] = \mu_k[j] \sum_{i=1}^{n} \phi(z_k^{(i)})$$

$$\mu_k[j] = \frac{\sum_{i=1}^{n} \phi(z_k^{(i)}) x^{(i)}[j]}{N_k}$$

Thus:

$$\overline{\boldsymbol{\mu}}_k = \frac{\sum_{i=1}^{n} \phi(z_k^{(i)}) \mathbf{x}^{(i)}}{N_k}$$

These match the given expressions, completing the solution.

1.2. Confusion Matrix

1. Two-Class Confusion Matrix

Accuracy

Accuracy is the ratio of correctly predicted instances to the total number of instances:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Substituting the values:

$$\text{Accuracy} = \frac{20 + 15}{20 + 15 + 10 + 5} = \frac{35}{50} = 0.7$$

Precision

Precision is the ratio of true positives to the total predicted positives:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{Precision} = \frac{20}{20 + 10} = \frac{20}{30} = \frac{2}{3} \approx 0.6667$$

Recall

Recall is the ratio of true positives to the total actual positives:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{Recall} = \frac{20}{20 + 5} = \frac{20}{25} = 0.8$$

F1-Score

Using the given formula:

$$\text{F1-score} = \frac{2}{\text{Recall}^{-1} + \text{Precision}^{-1}} = \frac{2}{1.25 + 1.5} = \frac{2}{2.75} = \frac{8}{11}$$

2. Multi-Class Confusion Matrix

Task 1: TP, FP, FN, TN for Class b

In a multi-class setting, for a specific class (e.g., class b), use a one-vs-rest approach:

- **True Positives (TP_b)**: Instances where actual is b and predicted is b. From the matrix, this is the cell at row b, column b:

$$\text{TP}_b = 540$$

- **False Positives (FP_b)**: Instances predicted as b but actual is not b. Sum of column b, excluding the TP_b:

$$\text{FP}_b = 23\,(\text{a} \to \text{b}) + 96\,(\text{c} \to \text{b}) + 2\,(\text{d} \to \text{b}) = 121$$

- **False Negatives (FN_b)**: Instances where actual is b but predicted is not b. Sum of row b, excluding TP_b:

$$\text{FN}_b = 10\,(\text{b} \to \text{a}) + 21\,(\text{b} \to \text{c}) + 14\,(\text{b} \to \text{d}) = 45$$

- **True Negatives (TN_b)**: Instances where actual is not b and predicted is not b. Sum of all elements not in row b and not in column b (i.e., the 3×3 submatrix of classes a, c, d and predictions a, c, d):

$$\text{TN}_b = 5 + 17 + 17 + 166 + 436 + 110 + 1 + 5 + 87 = 844$$

Task 2: Accuracy, Macro Precision, Macro Recall, Balanced Accuracy

- **Accuracy**

Accuracy is the ratio of correctly predicted instances (sum of diagonal elements) to the total instances:

$$\text{Correct Predictions} = 5 + 540 + 436 + 87 = 1068$$

$$\text{Total} = 1550$$

$$\text{Accuracy} = \frac{1068}{1550} \approx 0.6890$$

- **Macro Average Precision**

$$\text{MacroPrecision} = \frac{\sum_{k=1}^{K} \frac{\text{TP}_k}{\text{TP}_k + \text{FP}_k}}{K}, \quad K = 4$$

Compute for each class:

- Class a: $\text{TP}_a = 5$, $\text{FP}_a = 10 + 166 + 1 = 177$

$$\text{Precision}_a = \frac{5}{5 + 177} = \frac{5}{182} \approx 0.0275$$

- Class b: $\text{TP}_b = 540$, $\text{FP}_b = 121$

$$\text{Precision}_b = \frac{540}{540 + 121} = \frac{540}{661} \approx 0.8169$$

- Class c: $\text{TP}_c = 436$, $\text{FP}_c = 17 + 21 + 5 = 43$

$$\text{Precision}_c = \frac{436}{436 + 43} = \frac{436}{479} \approx 0.9102$$

- Class d: $\text{TP}_d = 87$, $\text{FP}_d = 17 + 14 + 110 = 141$

$$\text{Precision}_d = \frac{87}{87 + 141} = \frac{87}{228} \approx 0.3816$$

$$\text{MacroPrecision} = \frac{1}{4}(0.0275 + 0.8169 + 0.9102 + 0.3816) \approx 0.5341$$

- **Macro Average Recall**

$$\text{MacroRecall} = \frac{\sum_{k=1}^{K} \frac{\text{TP}_k}{\text{TP}_k + \text{FN}_k}}{K}$$

- Class a: $\text{FN}_a = 23 + 17 + 17 = 57$

$$\text{Recall}_a = \frac{5}{5 + 57} = \frac{5}{62} \approx 0.0806$$

- Class b: $\text{FN}_b = 45$

$$\text{Recall}_b = \frac{540}{540 + 45} = \frac{540}{585} = \frac{12}{13} \approx 0.9231$$

- Class c: $\text{FN}_c = 166 + 96 + 110 = 372$

$$\text{Recall}_c = \frac{436}{436 + 372} = \frac{436}{808} \approx 0.5396$$

- Class d: $\text{FN}_d = 1 + 2 + 5 = 8$

$$\text{Recall}_d = \frac{87}{87+8} = \frac{87}{95} \approx 0.9158$$

$$\text{MacroRecall} = \frac{1}{4}(0.0806 + 0.9231 + 0.5396 + 0.9158) = \approx 0.6148$$

- **Balanced Accuracy**

$$\text{BalancedAccuracy} = \frac{\sum_{k=1}^{K} \frac{\text{TP}_k}{\text{Total}_{\text{row}_k}}}{K}$$

Since $\text{Total}_{\text{row}_k} = \text{TP}_k + \text{FN}_k$, this is the recall for each class. Thus:

$$\text{BalancedAccuracy} = \text{MacroRecall} \approx 0.6148$$

Is Balanced Accuracy the Same as Accuracy? Why?

- **Accuracy ≈ 0.6890**
- **Balanced Accuracy ≈ 0.6148**

They are **not the same**. The reason is **class imbalance**:

- Class sizes: a (62), b (585), c (808), d (95). Total = 1550.
- Class c (808 instances) and class b (585) dominate, while class a (62) is small.
- **Accuracy** reflects overall performance, skewed toward larger classes (e.g., high TP for b and c boosts it).
- **Balanced Accuracy** averages recall per class equally, penalizing poor performance on smaller classes (e.g., recall_a = 0.0806 is very low).

Thus, due to imbalance, balanced accuracy is lower than accuracy.